

MATHS & SCIENCE OLYMPIAD

Foundation Session — Detailed Lesson Guide

Planrs Academy

Subject	Mathematics & Science	Level	Class 1–5 (Olympiad Beginner)
Duration	60–75 minutes	Guide	Arya the Explorer
Theme	Indian culture & everyday life	Motto	Every question is a new adventure!

■ Introduction & Learning Objectives

This foundation session welcomes young learners into the exciting world of Olympiad competitions through relatable Indian contexts — Diwali diyas, kolam patterns, temple bells, and monsoon rains. The session is structured into a **Maths Block** and a **Science Block**, each featuring concept teaching, interactive activities, and practice questions.

Learning Objectives

- Identify and extend number patterns using skip counting (2s, 5s, 10s).
- Recognise and classify basic 2D shapes in real-life Indian objects.
- Distinguish living things from non-living things using observable characteristics.
- Name and identify the five sense organs and relate each to everyday experiences.
- Identify India's four major seasons with associated natural events and festivals.
- Apply multi-step arithmetic to solve word problems within 30 seconds.

Catchphrase to reinforce throughout the session: "Every question is a new adventure!"

■ MATHS BLOCK

Topic 1 — Number Patterns

Arya decorates **Diwali diyas** in a pattern: **2, 4, 6, 8 ...** The story context uses familiar festivals to introduce skip counting naturally.

Concept Explanation

Skip counting means jumping forward by the same number each time.

Counting by 2s → 2, 4, 6, 8, 10, 12 ... (even numbers)

Counting by 5s → 5, 10, 15, 20, 25 ... (used for kolam dots & temple steps)

Counting by 10s → 10, 20, 30, 40 ... (money, bundles)

KEY RULE: Find the gap between any two consecutive numbers — that gap is the 'skip'.

Real-Life Indian Examples

Kolam dots are arranged in groups of 5 forming the pattern **5 → 10 → 15 → 20**. Temple steps are numbered **5 → 10 → 15 → 20 → 25**. These vivid examples help students visualise abstract sequences.

Classroom Activity — Pattern Hunt

Instructions

1. Ask students to clap on every 2nd count while the class counts aloud 1–20.
2. Display a 5×5 grid of diyas on the board; students circle every 2nd diya.
3. Have students draw their own kolam pattern using multiples of 5.
4. Discuss: What is the next number? How do you know?

Topic 2 — Shapes Around Us

Shapes are explored through **everyday Indian objects** to make geometry tangible and memorable for young learners.

Shape	Properties	Indian Example	Sides / Corners
Circle	Perfectly round, no corners	Roti, Rupee coin	0 sides, 0 corners

Triangle	3 straight sides, 3 corners	Traffic sign board	3 sides, 3 corners
Rectangle	4 sides; opposite sides equal	Window, door	4 sides, 4 corners
Square	4 equal sides, 4 corners	Handkerchief	4 sides, 4 corners

Cricket Pitch Connection

The **cricket pitch boundary** is an oval/ellipse — conceptually closest to a **circle** among the options given (square, triangle, circle, rectangle). This sports connection makes the shape question memorable and culturally relevant.

Maths Olympiad Practice Questions

Q#	Question	Answer
Q1	Complete the pattern: 5, 10, 15, 20, ____? A) 22 B) 25 C) 30 D) 24	B) 25 (skip +5 each time)
Q2	Which object is shaped like a cricket pitch boundary? A) Square B) Triangle C) Circle D) Rectangle	C) Circle
Q3	Who has more marbles? Arjun = 34, Priya = 41.	B) Priya (41 > 34)

■ SCIENCE BLOCK

Topic 3 — Living & Non-Living Things

Arya visits an Indian market and categorises objects around her — helping students apply scientific thinking to a familiar, bustling environment.

Characteristic	Living Things	Non-Living Things
Growth	Grow over time	Do not grow
Breathing / Respiration	Take in oxygen	No respiration
Nutrition / Eating	Need food for energy	Need no food
Reproduction	Produce offspring / seeds	Cannot reproduce
Response to stimuli	React to light, touch, sound	Do not react
Examples	Cow, Neem tree, Butterfly, Mango tree, Dog	Bullock cart, Stone idol, Pot, Cup, Wooden bench

Brain Break Activity — Stand Up / Sit Down

Read out each item. Students STAND if it is LIVING, SIT if NON-LIVING.

■ Cow — LIVING | ■ Stone — NON-LIVING | ■ Butterfly — LIVING

■ Wooden Bench — NON-LIVING | ■ Neem Tree — LIVING

Note: A wooden bench was once a living tree, but as a manufactured object it no longer grows, breathes, or reproduces — so it is classified as NON-LIVING.

Topic 4 — The Five Senses

Students explore sensory experiences through an **Indian morning routine**, connecting abstract biology to the richness of everyday life.

Sense Organ	Sense	Indian Morning Example	Key Vocabulary
Nose	Smell	Hot chai brewing	Aroma, fragrance, odour
Ears	Hearing	Temple bells ringing	Sound, vibration, loud/soft
Tongue	Taste	Guava with salt	Sweet, sour, salty, bitter

Skin	Touch / Feel	Soft woollen shawl	Rough, smooth, hot, cold
Eyes	Sight	Bright Holi colours	Colour, shape, bright/dim

Topic 5 — Indian Seasons

Season	Key Features	Animal / Plant Signs	Festival / Event
■ Summer	Hot afternoons, dry weather, school holidays	Mango fruit ripens, many birds migrate	Ugadi, Vishu, Baisakhi
■ Monsoon	Heavy rain, umbrellas, puddles, humidity	Frogs jump, earthworms emerge, plants sprout	Onam, Teej, Eid al-Adha
❄ Winter	Cool/cold, fog in north India, sweaters	Birds migrate south; bears hibernate elsewhere	Lohri, Pongal, Christmas
■ Autumn	Harvest time, mild weather, falling leaves	Crops ripen, migratory birds arrive	Navratri, Dussehra, Diwali

Science Olympiad Practice Questions

Q#	Question	Answer
Q1	From this list — Dog, Pot, Mango tree, Cup, Butterfly — how many are living things?	3 (Dog, Mango tree, Butterfly)
Q2	Meena smells biryani cooking. Which sense organ is she using?	Nose
Q3	Which season has frogs jumping and puddles everywhere?	Monsoon

■ Olympiad Champion Moment Challenge

This rapid-fire word problem simulates real Olympiad time pressure. Students must apply **multi-step arithmetic** within 30 seconds.

Problem: 12 birds sit on a tree. 5 more arrive. Then 3 fly away. How many birds are on the tree now?

Step 1: $12 + 5 = 17$ birds after arrivals. Step 2: $17 - 3 = 14$ birds after 3 fly away.

Strategy tip: Underline key numbers in the question. Write each step clearly. Check: does your answer make sense?

■ Session Recap — What We Learned Today

■ Number Patterns	Skip counting by 2s, 5s, and 10s reveals patterns quickly. Find the common gap and apply it to predict the next number.
■ Shapes in Daily Life	Circles (roti/coin), Triangles (traffic signs), Rectangles (windows), Squares (handkerchiefs) — geometry is everywhere around us.
■ Living vs Non-Living	Living things GROW, BREATHE, EAT, and REPRODUCE. Non-living things do none of these things.
■ Five Senses	Nose → smell, Ears → hearing, Tongue → taste, Skin → touch, Eyes → sight. Each sense organ has a dedicated function.
■ Indian Seasons	Summer (mangoes), Monsoon (frogs/puddles), Winter (Lohri/fog), Autumn (harvest/Diwali) — seasons drive nature and festivals.

■ Homework & Extension Activities

Take-Home Tasks

1. MATHS: Write the next 5 numbers in each sequence: 3, 6, 9, 12, ____ and 10, 20, 30, ____
2. SHAPES: Walk around your home. Draw 2 objects for each shape: circle, square, rectangle, triangle.

3. SCIENCE: Make a table with 5 living things and 5 non-living things you can see from your window.
4. SENSES: During dinner, write down one thing you noticed with EACH of your five senses.
5. CHALLENGE: Create your own 3-step word problem about animals or fruits and solve it.

Tips for Parents & Guardians

How to Support Your Young Olympiad Champion

- Revisit the session's catchphrase daily: "Every question is a new adventure!"
- Play skip-counting games during car rides (count lamp posts by 5s, etc.).
- Ask your child to identify shapes and living/non-living things while shopping.
- Encourage curiosity — reward thoughtful questions, not just correct answers.
- A downloadable practice worksheet is available from Planrs Academy to reinforce today's learning.